

20260420EB_MN (NCT, RNA)

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2026年4月20日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2026年4月22日 星期三

T2 -> T3

2026年4月24日 星期五

T3 -> T4 (~14 mL for over-weekend)

2026年4月27日 星期一

T4 -> T5

2026年4月29日 星期三

T5 -> T6

2026年5月1日 星期五

T6 -> T6 (~14mL for over-weekend)

2026年5月4日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info ^

	A	B	C
1	cell	passage	well#
2	WT11	P13	2
3	iPSC1000	P3+9	2
4	iPSC1066	P6+9	2
5	iPSC1108	P6+9	2

Table1 ^

	A	B	C	D
1	WT11		1000, 1066, 1108	
2		volume (mL)		volume (mL)
3	trypsin	3	trypsin	2.5
4	HI-FBS/CTWM	6	HI-FBS/CTWM	5

Table3 ✓

	A	B	C	D	E	F	G	H	I	J
1		conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	confluency the day after
2	WT11	1.44		5	7.2	A	24	300	24	50
3	iPSC1000	3.89	100	2	7.78	B	6	3890	2	60
4	iPSC1066	4.06	99	2	8.12	B	6	2030	4	40
5	iPSC1108	2.1	100	2	4.2	C	12	1400	3	60

Plate A ^

	1	2	3	4	5	6
A	110 (36uL virus in total, 12uL/4well)					
B						
C						
D	111 (45uL virus in total, 15uL/4well)					

Plate B ^

	1	2	3
A	iPSC1000	iPSC1066	
B			

Plate C ^

	1	2	3	4
A	iPSC1108			
B				
C				

2026年5月5日 星期二

Change T7 -> T7 if added virus

2026年5月6日 星期三

T7-> T7 if not added virus

2026年5月7日 星期四

T7 -> T8

2026年5月9日 星期六

1/2 T8 -> 1/2 T8

2026年5月11日 星期一

1/2 T8 -> 1/2 T8

2026年5月13日 星期三

1/2 T8 -> 1/2 T8

Imaging: NCT with MMS

- ✓ Prepare 50*2 2.5mM gluc NBA + T8 supp + 1mM pyruvate (I only add glucose to one tube)
- ✓ SirDNA 1:6000 -> 1 in 20uL --> 1000/6000*20=3.33uL
- ✓ Dilute MMS 3uL in 297uL media
- ✓ Prepare different concentration of MMS
- ✓ Mix well
- ✓ Take 1mL media with different MMS concentration, add 3.33uL 1/20SirDNA into it
- ✓ Mix well

PlateA							^
	1	2	3	4	5	6	
A	A	B	C	A'	B'	C'	
B	D	E	F	D'	E'	F'	
C	A	B	C	A'	B'	C'	
D	D	E	F	D'	E'	F'	

condition									/
	A	B	C	D	E	F	G	H	
1	condition #	MMS	media		condition #	MMS	treatment	media	
2	A	MMS 0mM	with 1mM pyruvate		A'	MMS 0mM	olaparib 200nM	with 1mM pyruvate	
3	B	MMS 0.5mM	with 1mM pyruvate		B'	MMS 0.5mM	olaparib 200nM	with 1mM pyruvate	
4	C	MMS 1mM	with 1mM pyruvate		C'	MMS 1mM	olaparib 200nM	with 1mM pyruvate	
5	D	MMS 1.5mM	with 1mM pyruvate		D'	MMS 1.5mM	olaparib 200nM	with 1mM pyruvate	
6	E	MMS 2mM	with 1mM pyruvate		E'	MMS 2mM	olaparib 200nM	with 1mM pyruvate	
7	F	MMS 2.5mM	with 1mM pyruvate		F'	MMS 2.5mM	olaparib 200nM	with 1mM pyruvate	

prepare media with different conc. of glucose with pyruvate							
	A	B	C	D	E	F	G
1	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume	100mM pyruvate (uL)
2	1100	2.5	440	50000	113.6	49886.4	500
3			total media volume	50.0	mL		

drug calculation on 5/13													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1			stock (mM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution (mM)	working (mM)	dilution factor from 1st dil.	total volume (mL)	volume of 1st dil.	volume of media
2	MM S	A							0		22000	0	1000
3		B	11800	100	300	3	297	118	0.5	236	2000	8.5	1991.5
4		C	11800	100	300	3	297	118	1	118	2000	16.9	1983.1
5		D	11800	100	300	3	297	118	1.5	78.6666666667	10000	127.1	9872.9
6		E	11800	100	300	3	297	118	2	59	2000	33.9	1966.1
7		F	11800	100	300	3	297	118	2.5	47.2	2000	42.4	1957.6
8		F	11800	100	300	3	297	118	2.5	47.2	500	10.6	489.4
9										total	40000	228.8	

time table for NCT treatment_MMS															
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	NCT														
2	condition#		condition	media	pyruvate	time	imaging	washout with 2.5mM gluc	note	recovery observation at 11:30pm, 5/13	recover imaging time	recovery time		observation at ~3pm on 5/14	
3	on 5/13/2026		110												
4	A	A1	MMS 0mM	2.5mM gluc	1mM	14:45	15:35	16:05		cells looked great and normal	11:32	~7.5h		alive	16:00
5	B	A2	MMS 0.5mM	2.5mM gluc	1mM	15:05	16:08	16:38		Many cells (~50%) were rounded up	11:38	~7h		a part alive	15:48
6	C	A3	MMS 1mM	2.5mM gluc	1mM	15:32	16:44	17:10	delayed >5min	Almost 100% died. Saw some survived non-MNs	11:44	~6.5h			15:54
7	D	B1	MMS 1.5mM	2.5mM gluc	1mM	16:05	17:14	17:45	delayed >5min	Didn't see any cells survived (didn't image)		~6h			
8	E	B2	MMS 2mM	2.5mM gluc	1mM	16:38	17:47	18:10	delayed >5min	Didn't see any cells survived (didn't image)		~6h			
9	F	B3	MMS 2.5mM	2.5mM gluc	1mM	17:10	18:20	18:38	delayed >5min	Saw very few cells not totally dead. Not sure if it's non-MN or not	11:50	~5h			died
10			111												
11	A	C 1	MMS 0mM	2.5mM gluc	1mM	19:00	20:01	20:30		cells looked great and normal	11:53	3.5h		alive	
12	B	C 2	MMS 0.5mM	2.5mM gluc	1mM	19:30	20:43	21:00	delayed >5min Due to some focus problem with specific positions	Most cells were fine, but some cells (<20%, mostly locate at upper right corner) started to round up	11:58	3h		alive	
13	C	C 3	MMS 1mM	2.5mM gluc	1mM	19:59	21:09	21:35	delayed >5min	Most cells were fine, but some cells (~20-30%, mostly locate at upper right corner) started to round up	12:04	2.5h		alive	
14	D	D 1	MMS 1.5mM	2.5mM gluc	1mM	20:30	21:40	21:55	delayed >5min	Most cells looked sick (bright and rounding up). ~30-40% died	12:12	2h	in general, the upper right corner would died the most and the upper left corner will survived the most. I change media from the upper left	alive	
15	E	D 2	MMS 2mM	2.5mM gluc	1mM	21:00	21:14	22:30	delayed >5min Due to some focus problem with specific positions	Many cells looked sick, ~20% died	12:19	~2h		alive	
16	F	D 3	MMS 2.5mM	2.5mM gluc	1mM	21:35	22:44	23:10		Most cells looked sick and started to round up	12:26	~1h		sick/dead	

NCT_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	
3	640	16	10	100 ms	
4	488	16	50	1s	

fileID_treatment_NCT							
	A	B	C	D	E	F	G
1	image	reporter	treatment	drug	time	pyruvate	
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	0mM MMS	NT	1h	yes	
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	NT	1h	yes	
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	0.5mM MMS	NT	1h	yes	
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	0.5mM MMS	NT	1h	yes	XY5 has shifted focus
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	1mM MMS	NT	1h	yes	
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	1mM MMS	NT	1h	yes	XY5 has shifted focus
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	1.5mM MMS	NT	1h	yes	
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	1.5mM MMS	NT	1h	yes	XY6 is sllightly out of focus XY7 is out of focus very often XY8 is out of focus for at frame 22
10	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	2mM MMS	NT	1h	yes	
11	com_110_MN_no-glu_50%_1s_30s_60x_0009	110	2mM MMS	NT	1h	yes	
12	com_110_MN_no-glu_50%_1s_30s_60x_0010	110	2.5mM MMS	NT	1h	yes	
13	com_110_MN_no-glu_50%_1s_30s_60x_0011	110	2.5mM MMS	NT	1h	yes	
14	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	0mM MMS	NT	1h	yes	
15	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	NT	1h	yes	
16	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	0.5mM MMS	NT	1h	yes	
17	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	0.5mM MMS	NT	1h	yes	XY2 has shifted focus
18	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	1mM MMS	NT	1h	yes	
19	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	1mM MMS	NT	1h	yes	
20	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	1.5mM MMS	NT	1h	yes	
21	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	1.5mM MMS	NT	1h	yes	3
22	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	2mM MMS	NT	1h	yes	
23	com_111_MN_no-glu_50%_1s_30s_60x_0009	111	2mM MMS	NT	1h	yes	XY2 has shifted focus
24	com_111_MN_no-glu_50%_1s_30s_60x_0010	111	2.5mM MMS	NT	1h	yes	
25	com_111_MN_no-glu_50%_1s_30s_60x_0011	111	2.5mM MMS	NT	1h	yes	Some frames of XY1 and XY2 have shifted focus

After recovery, image the SirDNA at 11:30pm

2026年5月14日 星期四

Check cell viability by bright field since PI is also in red fluorescence

I took the SirDNA image again with old confocal.
I found the cells with import reporter is more sensitive to MMS....
I checked the SirDNA. With import reporter, cells only survived with 0.5mM MMS. However, with export importer, they are still alive with 2mM.....
Hypothesis:
1. Import reporter is more toxic so the import rate is affected easier?
2. MMS is unstable in aqueous. (I googled it and found MMS can be hydrolyzed in water and lost its alkylating ability)

2026年5月15日 星期五

Imaging: NCT with MMS+PARP1 inhibitor

Because MMS is unstable in aqueous with high temperature, I prepared 1/100 primary dilution stock for each batch (i.e. import, export, RNA) and put it on ice. I also prepared 500uL media aliquot and warmed it up in 37C. I added 1/100 MMS right before adding it to cells (during I imaging the previous batch.)

1. Prepare 50*2 2.5mM gluc NBA + T8 supp + 1mM pyruvate
2. SirDNA 1:6000 -> 1 in 20uL --> 1000/6000*20=3.33uL
3. Dilute olaparib (10mM stock) 1:10
4. Take 32mL media and add 6.4uL 1mM olaparib
5. Dilute MMS 3uL in 297uL media
6. Prepare different concentration of MMS
7. Mix well
8. Take media without olaparib to prepare 1.5mM MMS for RNA exp
9. Mix well
10. Take 1mL media with different MMS concentration, add 3.33uL 1/20SirDNA into it
11. Mix well

prepare media with different conc. of glucose with pyruvate1							
	A	B	C	D	E	F	G
1	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume	100mM pyruvate (uL)
2	1100	2.5	440	50000	113.6	49886.4	500
3			total media volume	50.0	mL		

drug calculation on 5/1																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1											with olaparib (200nM)			without olaparib		
2			stock (mM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution (mM)	working (mM)	dilution factor from 1st dil.	total volume	volume of 1st dil.	volume of media	total volume	volume of 1st dil.	volume of media
3	MMS	A'							0		5000	0	1000	8000	0	1000
4		B'	11800	100	350	3.5	346.5	118	0.5	236	500	2.1	497.9			
5		C'	11800	100	350	3.5	346.5	118	1	118	500	4.2	495.8			
6		D'	11800	100	350	3.5	346.5	118	1.5	78.6666666667	500	6.4	493.6	10000	127.1	9872.9
7		E'	11800	100	350	3.5	346.5	118	2	59	500	8.5	491.5			
8		F'	11800	100	350	3.5	346.5	118	2.5	47.2	500	10.6	489.4			
9			stock (mM)						working (uM)		7500	31.8		37500		
10	olapari b		10	10	10	1	9	1	0.2	5000	19500	3.9	19496.1			

drug calculation on 5/15																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1											with olaparib (200nM)			without olaparib		
2			stock (mM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution (mM)	working (mM)	dilution factor from 1st dil.	total volume	volume of 1st dil.	volume of media	total volume	volume of 1st dil.	volume of media
3	MMS	A'							0		5000	0	1000	8000	0	1000
4		B'	11800	100	350	3.5	346.5	118	0.5	236	2000	8.5	1991.5			
5		C'	11800	100	350	3.5	346.5	118	1	118	2000	16.9	1983.1			
6		D'	11800	100	350	3.5	346.5	118	1.5	78.6666666667	5000	63.6	4936.4	10000	127.1	9872.9
7		E'	11800	100	350	3.5	346.5	118	2	59	2000	33.9	1966.1			
8		F'	11800	100	350	3.5	346.5	118	2.5	47.2	2000	42.4	1957.6			
9			stock (mM)						working (uM)		18000	292.4		48000		
10	olapari b		10	10	10	1	9	1	0.2	5000	30000	6.0	29994.0			

time table for NCT treatment_MMS 5/15									
	A	B	C	D	E	F	G	H	I
1	NCT								
2	condition#		condition	drug	media	pyruvate	time	imaging	recover with olaparib only
3	on 5/15/2026		110						
4	A'	A4	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	12:30	13:30	13:48
5	B'	A5	MMS 0.5mM	olaparib 200nM	2.5mM gluc	1mM	13:10	14:10	14:30
6	C'	A6	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	13:50	14:50	15:07
7	D'	B4	MMS 1.5mM	olaparib 200nM	2.5mM gluc	1mM	14:30	15:30	15:50
8	E'	B5	MMS 2mM	olaparib 200nM	2.5mM gluc	1mM	15:10	16:10	16:30
9	F'	B6	MMS 2.5mM	olaparib 200nM	2.5mM gluc	1mM	15:50	16:50	17:15
10			111						
11	A'	C4	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	16:30	17:30	17:50
12	B'	C5	MMS 0.5mM	olaparib 200nM	2.5mM gluc	1mM	17:15	18:14	18:30
13	C'	C6	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	17:50	18:50	19:15
14	D'	D4	MMS 1.5mM	olaparib 200nM	2.5mM gluc	1mM	18:30	19:30	19:50
15	E'	D5	MMS 2mM	olaparib 200nM	2.5mM gluc	1mM	19:15	20:10	20:30
16	F'	D6	MMS 2.5mM	olaparib 200nM	2.5mM gluc	1mM	19:50	20:50	21:10

fileID_treatment_NCT_20260515_MN_110,111							
	A	B	C	D	E	F	G
1	image	reporter	treatment	drug	time	pyruvate	
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	0mM MMS	200nM olaparib	1h	yes	
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	200nM olaparib	1h	yes	
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	0.5mM MMS	200nM olaparib	1h	yes	
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	0.5mM MMS	200nM olaparib	1h	yes	
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	1mM MMS	200nM olaparib	1h	yes	
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	1mM MMS	200nM olaparib	1h	yes	
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	1.5mM MMS	200nM olaparib	1h	yes	
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	1.5mM MMS	200nM olaparib	1h	yes	
10	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	2mM MMS	200nM olaparib	1h	yes	
11	com_110_MN_no-glu_50%_1s_30s_60x_0009	110	2mM MMS	200nM olaparib	1h	yes	
12	com_110_MN_no-glu_50%_1s_30s_60x_0010	110	2.5mM MMS	200nM olaparib	1h	yes	
13	com_110_MN_no-glu_50%_1s_30s_60x_0011	110	2.5mM MMS	200nM olaparib	1h	yes	
14	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	0mM MMS	200nM olaparib	1h	yes	
15	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	200nM olaparib	1h	yes	XY3 has shifted focus
16	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	0.5mM MMS	200nM olaparib	1h	yes	
17	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	0.5mM MMS	200nM olaparib	1h	yes	
18	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	1mM MMS	200nM olaparib	1h	yes	
19	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	1mM MMS	200nM olaparib	1h	yes	
20	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	1.5mM MMS	200nM olaparib	1h	yes	
21	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	1.5mM MMS	200nM olaparib	1h	yes	3
22	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	2mM MMS	200nM olaparib	1h	yes	
23	com_111_MN_no-glu_50%_1s_30s_60x_0009	111	2mM MMS	200nM olaparib	1h	yes	
24	com_111_MN_no-glu_50%_1s_30s_60x_0010	111	2.5mM MMS	200nM olaparib	1h	yes	
25	com_111_MN_no-glu_50%_1s_30s_60x_0011	111	2.5mM MMS	200nM olaparib	1h	yes	

Then, do a 40X timelapse recovery experiment (21:10-21:40)

1. clean up the oil at the bottom
2. mark positions of each well
3. image every 30min for every wells (from 11:00 pm 5/15 - 9:00pm 5/16)

Recovery imaging: starting from 11pm

[illegible]

Prepare 3uM PI in PBS and add 500uL into the media to have 1.5uM PI at around 9:50pm on 5/16

20260516_MN_MMS_recovery_PI			
	A	B	C
1	well	time point	fileid_(DIC, 640, 561)
2	A4	22:12-22:30	MN_MMS_200nM-olaparib_40x_0000
3	A5		MN_MMS_200nM-olaparib_40x_0001
4	A6		MN_MMS_200nM-olaparib_40x_0002
5	B4		MN_MMS_200nM-olaparib_40x_0003
6	B5		MN_MMS_200nM-olaparib_40x_0004
7	B6		MN_MMS_200nM-olaparib_40x_0005
8	C4		MN_MMS_200nM-olaparib_40x_0006
9	C5		MN_MMS_200nM-olaparib_40x_0007
10	C6		MN_MMS_200nM-olaparib_40x_0008
11	D4		MN_MMS_200nM-olaparib_40x_0009
12	D5		MN_MMS_200nM-olaparib_40x_0010
13	D6		MN_MMS_200nM-olaparib_40x_0011

RNA extraction

Treat cells at 20:20-30; collect cells at 21:25 on 5/15 (MMS was prepared fresh)
Use 1mL for 1 well of 6-well plate; use 500uL for 1 well of 12-well plate

Well1			
	1	2	3
A	0mM MMS	0mM MMS	0mM MMS + 200nM olaparib
B	1.5mM MMS	1.5mM MMS	1.5mM MMS+ 200nM olaparib

Well2					^
	1	2	3	4	
A	0mM MMS				
B	1.5mM MMS				
C	1.5mM MMS				